



P38

China's AI Classification: Institutional & Technical Design — A Multi-Dimensional AI Classification Governance Platform for Education

中国的 AI 分级：制度与技术设计



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First said clearly in one sentence: AI classification is not labeling models "good/bad," but doing dynamic authorization for "**who in what scenario, using AI of what capability, doing what task, supervised by whom, using how much quota, in what time window.**" This paper translates China's top-down regulatory genealogy (PIPL → Minor Protection Regulations → Generative AI Measures → Labeling Measures + GB 45438 → Ministry of Education "classification and grading" policy) and international comparisons (EU AI Act

four-level risk, COPPA, GDPR-K) into a deployable ten-dimension classification code, PDP/PEP decision functions, ten-layer system architecture, data model and API, stage strategy matrix, and gives the essence and dross of technology selection, failure boundaries and falsifiable predictions. **The full text contains no new experiments; all design proposals marked "design proposal • to be implemented," quota/duration numbers marked "strategy example • configurable," external regulations and tools all with official sources and three-state marking.**

Evidence caliber (declared only once in full text): Chinese regulations checked to gov.cn / npc.gov.cn / cac.gov.cn / moe.gov.cn / std/openstd.samr.gov.cn official originals, access date 2026-09-21; international systems checked to European Commission, FTC/Federal Register, ICO, NIST, OASIS official pages; technical components checked to CNCF, NVIDIA, Meta HF, Microsoft, BerriAI official docs and arXiv. Three states:  verified (official original accessible) /  partially verified (existence confirmed but individual fields are guidance documents or second-hand retelling) /  no evidence found (marked [CITATION NEEDED]). UDOS's own measured numbers almost not cited in this paper; where cited all with verified/cpu-proto/unverified three-level marking.

Abstract

After generative AI enters education scenarios, both "one-size-fits-all ban" and "borderless opening" are unsustainable: the former deprives students of the opportunity to learn AI literacy, the latter exposes minors to multiple risks of content safety, data privacy, over-reliance and academic integrity. China's regulatory system has given the institutional direction of "classification and grading"—the Interim Measures for the Administration of Generative Artificial Intelligence Services (Order No. 15 of seven departments, effective 2023-08-15) Article 3 explicitly "classified and graded supervision," in 2026-04 the Ministry of Education and five units' "AI + Education" Action Plan formally proposed "classify and grade to determine safety protection standards," the 2025 Steering Committee's Guidelines for Generative AI Use by Primary and Secondary School Students clearly "at primary school stage prohibit students from independently using open-ended content generation functions." But institutional direction does not equal executable technical solutions.

This paper's core contribution is translating institutional direction into engineering design: ① building a **ten-dimension classification code** (safety S0–S4, age A0–A5, content topic C0–C4, cognitive readiness L0–L5, intelligence I0–I5, content source G0–G4, version V0–V3, permission P0–P5, quota Q0–Q5, time T0–T4), where the user's original list items 3/6 were both named "content grading," this paper rules to split into "content topic/risk grading" and "content source/originality grading"; ② defining the **decision function** $\text{Decision} = f(\text{identity, age, role, school stage, cognitive readiness, scenario, task, content, requested I level, version, permission, quota, time, guardian consent, teacher supervision})$, outputting nine enumerations, denial must give educational alternatives; ③ designing a **ten-layer system architecture** (access→identity organization→classification policy center→content safety→model gateway→education orchestration→memory learning state→quota time→audit compliance→admin backend), adopting PDP/PEP separation and Policy as Code; ④ giving **data model, key APIs, stage strategy matrix and education orchestration mechanisms** (pre-task gate, Socratic hints, process evidence, AI usage declaration, teacher final assessment, exam lock); ⑤ establishing mapping with UDOS governance system (L0–L5 autonomy, evidence engineering rules first, TransferBundle least privilege, BFT-lite multi-person review, quality saturation law); ⑥ doing **essence and dross** critique of OPA/Casbin, Llama Guard/NeMo Guardrails/Presidio/LiteLLM, AI detection tools, age assurance technology; ⑦ giving **failure boundaries, falsifiability conditions and predictions**.

Keywords: AI classification; education governance; generative artificial intelligence; Policy as Code; PDP/PEP; minor protection; content labeling; progressive opening; human in loop; auditable

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** after generative AI enters education scenarios, both "one-size-fits-all ban" and "borderless opening" are unsustainable; Chinese regulations have proposed "classification and grading" direction (Generative AI Measures Article 3, Ministry of Education 2026 Action Plan), but lacking translation from institution to executable technical solutions.
 - **Method/argument:** reconstructing "classification" from a single age switch to a ten-dimension orthogonal dynamic authorization system, using PDP/PEP separation + Policy as Code to decouple policy from models, using the education orchestration layer to ensure "think yourself first, AI as hint coach, teacher retains final judgment."
 - **Evidence threads:** (1) China's 14 regulations/policies checked to official originals (including Generative AI Measures Order No. 15 of seven departments, Minor Protection Regulations State Council Order No. 766, PIPL Article 31, Labeling Measures Guoxinban Tongzi [2025] No. 2 + GB 45438-2025, Ministry of Education 2026 "AI+Education" Action Plan); (2) international comparison checked to EU AI Act (Annex III education high risk, emotion recognition Art.5(1)(f) prohibited), COPPA 2025 revision, GDPR Art.8; (3) technical components checked to OPA/CNCF, NIST SP 800-162, Llama Guard 3, NeMo Guardrails, Presidio, LiteLLM, conformal prediction arXiv:2107.07511; (4) AI detection tool bias has peer-reviewed hard evidence Liang et al. 2023, Patterns 4(7) 100779.
 - **Contribution:** ten-dimension classification code system, decision function and output enumerations, ten-layer system architecture, data model and API, stage strategy matrix, UDOS governance mapping, technology selection essence/dross critique, failure boundaries and falsifiable predictions.
-

1 Problem and Positioning: Classification Is Not Labeling AI, It Is Authorizing "Person-Scenario-Capability"

One-sentence conclusion: AI governance failure in education scenarios is essentially treating "can AI be used" as a Boolean switch, while what truly needs governance is the multi-dimensional constraint set "who, in what scenario, using what capability, doing what task, supervised by whom."

1.1 Both Extremes Are Unsustainable

Current education attitudes toward generative AI show two poles:

- **"One-size-fits-all ban":** some schools/regions prohibit students from using any generative AI at school. Its cost is depriving students of the opportunity to learn AI literacy—while the 2024-12 Ministry of Education General Office Notice on Strengthening AI Education in Primary and Secondary Schools has clearly required advancing AI education by stage (lower primary perception experience, upper primary and junior high understanding and use, senior high project creation and advanced technology application), banning contradicts policy direction.
- **"Borderless opening":** some platforms directly open full-function generative AI to minors, without age determination, content adaptation, time limits or process logging. This directly violates the

Minor Online Protection Regulations (State Council Order No. 766, effective 2024-01-01) required "time management, permission management, consumption management," also violates Generative AI Measures Article 10 "prevent minors from over-reliance and addiction."

The feasible path between the two poles is **graded governance**—but "grading" is often simplified to "one-size-fits-all by age," which is equally insufficient.

1.2 Why Single Age Grading Is Insufficient

Single age grading has at least four blind spots:

1. **Same age, cognitive readiness varies greatly:** same 12 years old, some students can critically use AI, others still need full teacher proxy. Age is a threshold, not a capability proof.
2. **Same age, scenarios vary greatly:** using AI for Socratic hints in classwork is learning, using AI to generate answers in exams is cheating. Scenarios must be graded independently of age.
3. **Same age, task cognitive complexity varies greatly:** using AI to explain a concept (L1 understanding) and using AI to write a research paper (L5 creation) need completely different permissions and supervision.
4. **Same age, AI autonomy varies greatly:** using AI for retrieval Q&A (I1) and letting AI as a multi-agent autonomously execute tasks (I5) have completely different risk levels.

Therefore, this paper reconstructs "grading" as a **ten-dimension orthogonal dynamic authorization system**.

1.3 This Paper's Positioning and Boundaries

This paper is an **institutional + technical design paper**, not an empirical research paper. The question it answers: under China's current regulatory framework, how should an AI classification governance platform for education scenarios be designed? It does not claim to have implemented or verified the platform—all architectures, interfaces, strategy matrices are **design proposals • to be implemented**, marked [RESULT NEEDED].

This paper does not discuss: technical details of general AI safety alignment (RLHF, red-team training etc.), capability evaluation of specific models, empirical learning-effect studies in educational psychology. These are adjacent but independent topics.

Chapter summary: AI classification's governance object is not AI itself, but the multi-dimensional combination "person-scenario-capability-task-supervision"; single age switch has four blind spots, needing a ten-dimension orthogonal system.

2 Institutional Basis: China's Regulatory Genealogy and International Comparison

One-sentence conclusion: China has formed a top-down regulatory genealogy "law (PIPL) → administrative regulation (Minor Protection Regulations) → departmental rules (Generative AI Measures/algorithm/deep synthesis/labeling/children provisions) → national standard (GB 45438) → education industry policy (Ministry of Education classification and grading)," "classification and grading" has risen from a regulatory principle in the cyberspace field to a formal policy expression in education; in international comparison the EU AI Act explicitly lists education admission/scoring/exam monitoring as

high risk, emotion recognition directly prohibited, providing a risk-grading reference frame for China's solution.

2.1 China's Regulatory Genealogy (by legal force level)

Table 1 China AI classification governance related regulatory policy genealogy (14 items, all checked to official sources, access date 2026-09-21)

Level	Regulation/ policy name	Issuing body	Document/ standard no.	Effective date	Core meaning for AI classification	Verification
Law	Personal Information Protection Law	NPC Standing Committee	President Order No. 91	2021-11-01	Article 31: under 14 need guardian consent + special rules; Article 28 sensitive personal information	✓
Administrative regulation	Minor Online Protection Regulations	State Council	State Council Order No. 766	2024-01-01	Requires time/ permission/ consumption management; anti-addiction/ anti-bullying chapters; covers all under-18	✓
Departmental rule	Interim Measures for Generative AI Services	Seven departments	Seven-department Order No. 15	2023-08-15	Article 3 "classified and graded supervision" ; Article 10 anti-addiction; Article 12 content labeling; Article 17 algorithm filing	✓
Departmental rule	Algorithm Recommendation Management Provisions	Four departments	Four-department Order No. 9	2022-03-01	Article 24 algorithm filing; minor non-addiction mode; turn off algorithm recommendation option	✓
Departmental rule	Deep Synthesis Management Provisions	Three departments	Three-department Order No. 12	2023-01-10	Deep synthesis content prominent labeling; classified graded filing	✓
Departmental rule	Children's Personal Information Online Protection Provisions	CAC	CAC Order No. 4	2019-10-01	"Children" = under 14; guardian consent; dedicated personnel; keep records	✓
Departmental rule	AI-Generated Synthetic Content Labeling Measures	Four departments	Guoxinban Tongzi [2025] No. 2	2025-09-01	Explicit labels + implicit labels (metadata/ watermark) mandatory; conversational	✓

Level	Regulation/ policy name	Issuing body	Document/ standard no.	Effective date	Core meaning for AI classification	Verification
					generation must prompt labels	
National standard	Cybersecurity Technology AI- Generated Synthetic Content Labeling Method	SAMR + SAC	GB 45438-2025	2025-09-01	Specifies explicit/ implicit label technical methods; non- compliant cannot serve domestic public	✓
Normative document	Education Informatization 2.0 Action Plan	MOE	Jiaoji [2018] No. 6	2018-04	Information literacy assessment, online learning space (policy source)	✓
Normative document	Notice on Strengthening AI Education in Primary and Secondary Schools	MOE General Office	[CITATION NEEDED] (doc no. not verbatim checked)	2024-12	By stage: lower primary perception→junior high understanding use→senior high project creation	●
Normative document	Opinions on Accelerating Education Digitalization	MOE and nine departments	[CITATION NEEDED]	2025-04	Implement algorithm and large model filing mechanisms; regulate AI entering campus	●
Normative document	"AI + Education" Action Plan	MOE and five units	[CITATION NEEDED]	2026-04	Article (16) "classify and grade to determine safety protection standards" ; education large model safety review	●
Guidance document	Guidelines for Generative AI Use by Primary and Secondary Students (2025 edition)	MOE Basic Education Teaching Steering Committee	no order no.	2025-05	"Primary stage prohibit students independently using open- ended content generation" ; junior high moderate exploration; senior high application	●

Level	Regulation/ policy name	Issuing body	Document/ standard no.	Effective date	Core meaning for AI classification	Verification
Analogical reference	Notice on Further Strict Management to Effectively Prevent Minors Addicted to Online Games	National Press and Publication Administration	Guoxin Chufa [2021] No. 14	2021-09-01	Only borrows "age+time+real- name" institutional logic; analogy not directly applicable to AI education	✓

Table 1 note: seven departments = CAC, NDRC, MOE, MOST, MIIT, MPS, NRTA; four departments (algorithm recommendation) = CAC, MIIT, MPS, SAMR; three departments (deep synthesis) = CAC, MIIT, MPS; four departments (labeling measures) = CAC, MIIT, MPS, NRTA. Fact correction: algorithm recommendation provisions by four departments not seven.

2.2 Three Institutional Main Lines of the Regulatory Genealogy

From Table 1 can extract three institutional main lines running through the genealogy, directly corresponding to this paper's technical design dimensions:

Main line one: classified and graded supervision—Generative AI Measures Article 3 establishes the "inclusive, prudent, classified and graded supervision" principle, MOE 2026 Action Plan lands it in education scenarios ("classify and grade to determine safety protection standards"). This is the direct legal source of this paper's "ten-dimension classification code."

Main line two: layered minor protection—PIPL Article 31 + children provisions set "under 14" as the hard threshold for guardian consent; Minor Protection Regulations extend protection scope to all under-18, requiring time/permission/consumption management. This means age grading cannot cut only at 14, but should cover the full A0–A5 and differentiate design (<14 strong guardian consent, 14–18 minor mode + school/parent informed).

Main line three: content labeling and traceability—deep synthesis provisions (2023) → labeling measures + GB 45438 (2025-09-01) form mandatory technical requirements of "explicit labels + implicit labels (metadata/digital watermark)." This is the technical legal source of this paper's "content source/ originality grading G0–G4" and "process evidence" mechanisms.

2.3 International Comparison: EU AI Act Four-Level Risk and Education Positioning

Table 2 EU AI Act four-level risk and education scenario comparison (checked to European Commission official pages, access date 2026-09-21)

Risk level	Definition points	Education scenario examples	Obligations
Unacceptable risk (Art.5)	Obvious threat to fundamental rights, directly prohibited	Emotion recognition AI in education institutions/workplaces (Art.5(1)(f) explicitly prohibited); social scoring; biometric classification inferring protected attributes	Prohibited from market placement/use
High risk (Art. 6+Annex III)	May significantly adversely affect health/safety or fundamental rights	Annex III category 3 "education and vocational training": admission/enrollment screening, learning outcome assessment, education level determination, exam process behavior monitoring	Full lifecycle compliance: risk management, data governance, technical docs, record keeping, transparency and human supervision, robustness and cybersecurity, conformity assessment, CE mark, registration
Limited risk (Art.50 transparency)	Not prohibited or high risk, but must fulfill transparency obligations	General education chatbots, AI tutoring dialogue (not for scoring/admission); AI-generated education materials	Inform users they are interacting with AI; disclose AI-generated/manipulated content
Minimal risk	Vast majority of AI applications, no specific obligations	Classwork assist suggestions, personalized practice recommendations	No mandatory obligations; encourage self-regulation and AI literacy

Table 2 note: EU AI Act (Regulation (EU) 2024/1689) effective 2024-08-01; prohibition clauses and AI literacy obligations applicable from 2025-02-02; GPAI model obligations from 2025-08-02; vast majority of high-risk obligations from 2026-08-02.

EU AI Act's three key implications for this paper:

1. **Education is not a single risk level:** AI for admission/scoring/exam monitoring is high risk, chatbots for class tutoring are limited risk—this corroborates this paper's "scenario independent grading" design.
2. **Emotion recognition in education scenarios directly prohibited:** Art.5(1)(f) explicitly prohibits using emotion recognition AI in education institutions and workplaces. This paper's "cognitive readiness L0–L5" must be strictly limited to "task cognitive complexity adaptation labels," **must not** move toward emotion recognition or mental state inference.
3. **High-risk obligations' core is "human supervision + record keeping":** this fully aligns with this paper's "human in loop, explainable auditable" design principles.

2.4 Other International System Points

- **COPPA (US):** children under 13 personal information must obtain verifiable parental consent (VPC); 2025-01-16 FTC passed final revision rules, adding knowledge-based authentication (KBA) and facial recognition (must human review) as VPC methods. Different age line from China PIPL Article 31 (<14), but institutional logic consistent—low-age strong guardian consent.
- **GDPR Art.8 (EU):** digital service consent age default 16, member states may lower within 13–16 (UK 13, Germany 16, France 15, Austria/Italy/Spain 14, latter three second-hand academic mapping [CITATION NEEDED]). Cross-border deployment needs strictest or per-country adaptation.
- **Age Assurance (UK ICO):** risk-graded age confidence—low risk can self-declare, high risk needs hard identifiers (government ID)/age estimation multi-factor. **Self-declared age easily bypassed**, facial

age estimation has bias and privacy controversy for minors, different skin colors. Education scenarios should prioritize government digital identity or parent account verification, AI age estimation only as auxiliary signal.

- **FERPA (US):** protects student education record privacy; whether inferential analysis AI vendors generate on their own servers without returning to school constitutes FERPA "education records," as of early 2025 DOE has no final clear interpretation 🟡.

Chapter summary: China has formed a top-down "classification and grading" regulatory genealogy, three main lines (classified graded supervision, layered minor protection, content labeling traceability) directly correspond to technical design dimensions; EU AI Act proves education scenarios need grading by use (high risk/limited risk), emotion recognition directly prohibited, high-risk obligations core is human supervision + record keeping.

3 Ten-Dimension Classification Code

One-sentence conclusion: the ten-dimension classification code turns "grading" from vague policy slogan into computable, versionable, auditable engineering objects; where the user's original list items 3/6 were both named "content grading," this paper rules to split into "content topic/risk grading (C0–C4)" and "content source/originality grading (G0–G4)," avoiding policy engine conflicts.



Figure 1 Ten-dimension classification overview: Decision=f(identity, age, role, scenario, content, task, model, version, permission, quota, time, consent, supervision). Ten dimensions orthogonal: safety is red line, age is threshold, cognition is adaptation, autonomy is routing, permission/quota/time are constraints, version is rollback. (Concept schematic)

3.1 Ten-Dimension Code Definition Master Table

Table 3 Ten-dimension classification code definition and education mapping

#	Dimension	Code	Level definitions	Education mapping	Technical implementation
1	Safety grading	S0–S4	S0 normal/S1 remind/S2 block/S3 high risk to human/S4 crisis intervention	Self-harm • violence • pornography • illegal • psychological crisis • privacy; S4 stops generation, to guardian/school psychologist	Multimodal moderation, classifiers, crisis intervention tickets
2	Age grading	A0–A5	A0 preschool(0-5)/A1 lower primary(6-8)/A2 upper primary(9-11)/A3 junior high(12-14)/A4 senior high(15-18)/A5 university adult(18+)	Low-age default no free generation, teacher/parent proxy; senior high progressive opening	IAM, student status, guardian consent, age assertions
3	Content topic/risk grading	C0–C4	C0 general/C1 subject child-appropriate/C2 controversial/C3 sensitive high risk/C4 adult professional	Labels include age-appropriateness, factuality, controversiality, source credibility; violence/porn/self-harm/privacy blocked or guided by level	Content labels, classifiers, knowledge base, RAG
4	Cognitive readiness	L0–L5	L0 remember/L1 understand/L2 apply/L3 analyze/L4 evaluate/L5 create (Bloom taxonomy)	User explicitly renamed "intelligence grading" to cognitive readiness ; only internal adaptation labels, prohibited from labeling/ranking/tracking/predicting future, appealable resettable	Task templates, stage assessment, dynamic update
5	Intelligence/autonomy grading	I0–I5	I0 no AI/I1 retrieval/I2 template single-turn/I3 multi-turn dialogue/I4 tool proxy/I5 multi-agent autonomous	Low-age limited I0–I2, senior high controlled I3–I4 pilot, teachers I4–I5; echoes UDOS L0–L5, SAE J3016	Model gateway, capability registration cards, tool whitelist
6	Content source/originality grading	G0–G4	G0 human original/G1 AI-assisted/G2 AI-generated reviewed/G3 AI direct/G4 AI	Watermark + originality detection + process records; supports academic integrity and teaching evaluation	Digital watermark, originality detection, process evidence base

#	Dimension	Code	Level definitions	Education mapping	Technical implementation
			ghostwritten undeclared		
7	Version grading	V0–V3	V0 experiment/V1 pilot/V2 stable default/V3 mandatory	Policy • model • label • SDK versions; canary/rollback/audit; education default stable version no auto upgrade	Version management, canary release, rollback mechanism
8	Permission grading	P0–P5	P0 none/P1 read-only/P2 limited dialogue/P3 generation/P4 API/tools/P5 admin	Roles: student/parent/teacher/research/admin/regulator; final_assessment evaluation right only teacher	RBAC+ABAC, OPA/Casbin
9	Quota grading	Q0–Q5	tokens/calls/duration/concurrency/cost/characters/images	Dynamic by age • stage • task • credit; low-age low quota, teachers higher	Quota center, rate limiting, metering
10	Time grading	T0–T4	T0 disabled/T1 class supervised/T2 home time/T3 free time/T4 exam lock	Time window/continuous duration/daily cumulative/cooldown/night lock/class mode; exam mode global disable	Clock policy, scenario mode, screen time

3.2 Ruling Note on Items 3/6 Naming Collision

In the user's original design material items 3 and 6 were both named "content grading," which would cause two same-name field conflicts in the policy engine. This paper rules as follows:

- **Item 3 = content topic/risk grading (C0–C4):** answers "what this content is about, how high the risk, whether age-appropriate." It is the content safety and age-adaptation dimension.
- **Item 6 = content source/originality grading (G0–G4):** answers "who wrote this content, how much AI participated, whether declared." It is the academic integrity and process traceability dimension, directly corresponding to Labeling Measures + GB 45438 explicit/implicit label requirements.

The "content presentation grading P0–P4 (plain text/image-text/multimodal/interactive/immersive)" proposed in another draft can serve as **alternative eleventh dimension**, but this paper does not include it in core ten—because presentation mode is more an interaction design choice than a governance decision dimension. If included, suggest naming "presentation modality grading M0–M4" to avoid confusion with C/G.

3.3 Three Key Distinctions

Distinction one: safety grading (S) vs content topic grading (C). S is red line—S3/S4 directly block or to human, not discussing "whether age-appropriate"; C is adaptation—C2 controversial content in

senior high can guide discussion, in primary blocked. S takes priority over C: any content first passes S red line, then does C adaptation.

Distinction two: cognitive readiness (L) vs intelligence grading (I). L is how much cognitive capability the **task itself** needs (Bloom taxonomy), determining AI interaction mode (L0–L1 direct feedback, L2–L3 hints, L4–L5 debate/counter-view); I is how much autonomy **the AI is allowed**, determining model routing and tool permissions. The two orthogonal: an L5 (create) task can use I2 (template single-turn) AI for hints, an L1 (understand) task should not use I5 (multi-agent autonomous) AI to execute.

Distinction three: age (A) vs cognitive readiness (L). Age is **legal and institutional threshold** (<14 needs guardian consent), cognitive readiness is **internal adaptation label**. Age cannot be skipped (legal hard constraint), cognitive readiness dynamically adjustable, appealable, resettable. **Strictly prohibited** from using cognitive readiness labels to rank, track or predict students' future—this is an ethical red line, also echoing EU AI Act strict constraints on education high-risk systems.

Chapter summary: ten-dimension code turns grading into computable engineering objects; items 3/6 naming collision ruled to split into content topic (C) and content source (G); three key distinctions (S vs C, L vs I, A vs L) ensure dimension orthogonality, policy no conflicts.

4 Decision Function and PDP/PEP Architecture

One-sentence conclusion: classification decision core is a pure function $\text{Decision} = f(\text{input attributes}) \rightarrow \text{output enumeration} + \text{constraints}$, adopting PDP (Policy Decision Point) and PEP (Policy Enforcement Point) separated architecture, policy decoupled from models, each decision explainable, versionable, auditable.

4.1 Decision Function Definition

$$\text{Decision} = f(\text{identity}, \text{age}, \text{role}, \text{school stage}, \text{cognitive readiness}, \text{scenario}, \text{task cognitive level}, \text{content safety/topic/source}, \text{requested I level}, \text{version}, \text{permission}, \text{quota}, \text{time}, \text{guardian consent}, \text{teacher supervision})$$

Inputs divided into four categories:

- **Subject attributes:** identity (userId), age (A), role, school stage, cognitive readiness (L), guardian consent status, teacher supervision status.
- **Scenario attributes:** scenario mode (class/homework/exam/home/project), device, location, network environment.
- **Resource attributes:** task type, task cognitive level (L), content safety level (S), content topic level (C), content source level (G), requested intelligence level (I).
- **Environment attributes:** version (V), permission (P), quota remaining (Q), time window (T), current clock.

4.2 Output Enumerations

Table 4 Decision output enumerations and meanings

Enumeration	Meaning	Typical trigger conditions
ALLOW	Allow, execute per default mode	All constraints satisfied
DENY	Deny, return educational alternative	Safety S3/S4, exam T4, low-age no permission, quota exhausted
ALLOW_WITH_GUARDRAILS	Allow but with guardrails (mandatory citation/process record/no direct answer)	Junior high generated essays, senior high projects
REQUIRE_SUPERVISION	Execute after teacher/parent supervision	Low-age requests high I level, sensitive content C2/C3
REQUIRE_VERIFICATION	Output after citation/verification/fact check	Controversial content, senior high research tasks
TEACHER_ONLY	Only teacher usable, student side routes to teacher proxy	Preschool/lower primary A0–A1 generative requests
DOWNGRADE_CAPABILITY	Downgrade to lower I level or more conservative model	Request I4 but policy only allows I2
PROXY_TEACHER	Teacher/parent proxy executes, student not directly interacting	Low-age special needs

Key design: denial is not cold "prohibition". When decision is DENY, the system must return an **educational alternative**, e.g.: "At this stage first write an outline yourself for 10 minutes, then I can help check logic and wording." This is the essential difference between education scenarios and general content moderation—the purpose of moderation is not preventing learning, but guiding learning.

4.3 Decision Input/Output Example

Input (JSON):

```
{
  "subject": {
    "user_id": "u123", "role": "student", "age": 10, "grade": "Primary Grade 4",
    "guardian_consent": true,
    "readiness": {"knowledge": 0.4, "self_control": 0.5, "risk_understanding": 0.3}
  },
  "scene": {"mode": "homework", "teacher_supervised": false, "exam_mode": false},
  "task": {"type": "writing", "subject": "Chinese", "cognitive_level": "L2"},
  "content": {"safety_level": "S0", "topic_level": "C1", "source_level": "G2"},
  "model": {"requested_level": "I3"},
  "quota": {"daily_remaining": 8},
  "time": {"current_window": "T2", "used_minutes": 12}
}
```

Output (JSON):

```
{
  "decision_id": "d_001",
  "allow": true,
  "mode": "SOCRATIC_HINT",
  "model_level": "I2",
  "permissions": ["chat.limited", "rag.textbook"],
  "constraints": [
    "No direct full essay",
    "First submit own outline",
    "Must give citations",
    "Record modification process"
  ],
  "quota": {"max_tokens": 1200, "remaining_calls": 8},
  "time": {"max_minutes": 15, "cooldown_minutes": 30},
  "human_in_loop": "parent_notify_if_sensitive",
  "policy_version": "v2.3.1",
  "model_version": "edu-llm-v1.8-stable",
  "reasons": ["age_A2_default_I2", "writing_task_require_outline_first", "quota_sufficient"]
}
```

Note the output contains `policy_version`, `model_version`, `reasons`—this is the basis of explainability and auditability. Each decision can be traced to "which policy, which model version, why decided so."

4.4 PDP/PEP Separation Architecture

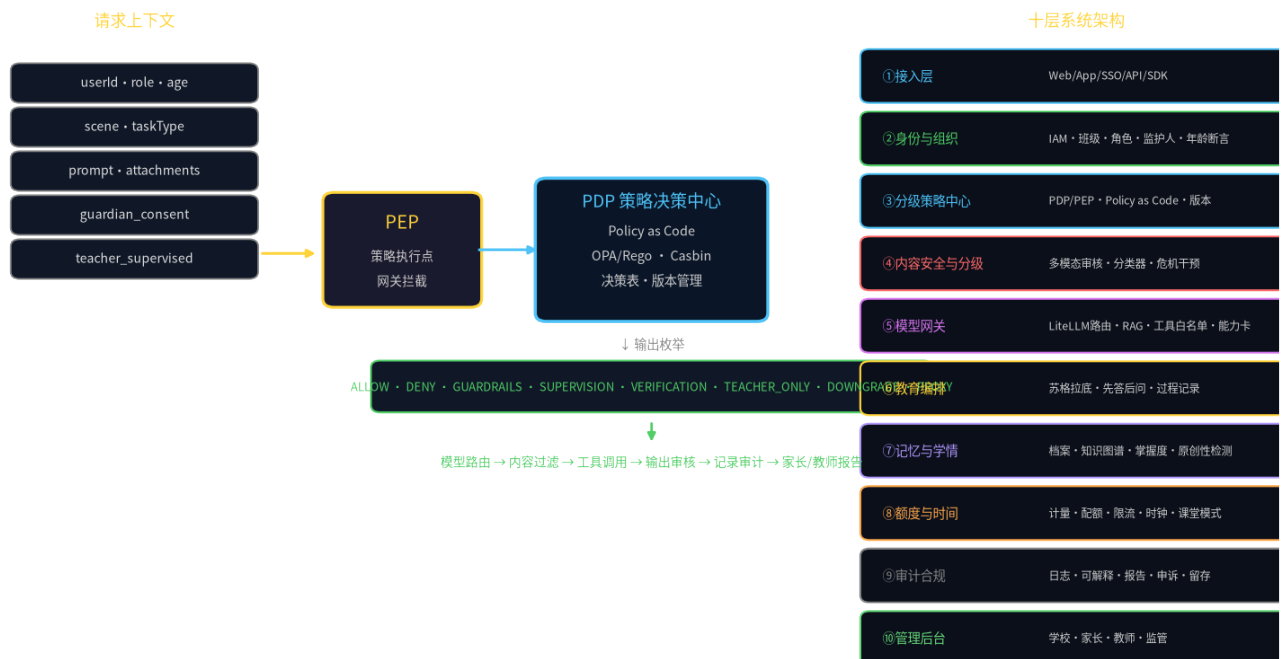


Figure 2 Decision flow (PEP→PDP→output enumeration) and ten-layer system architecture. PEP deployed at gateway/client/LMS entry, responsible for intercepting requests and executing decisions; PDP independently deployed, responsible for policy evaluation, decoupled from business code. (Concept schematic)

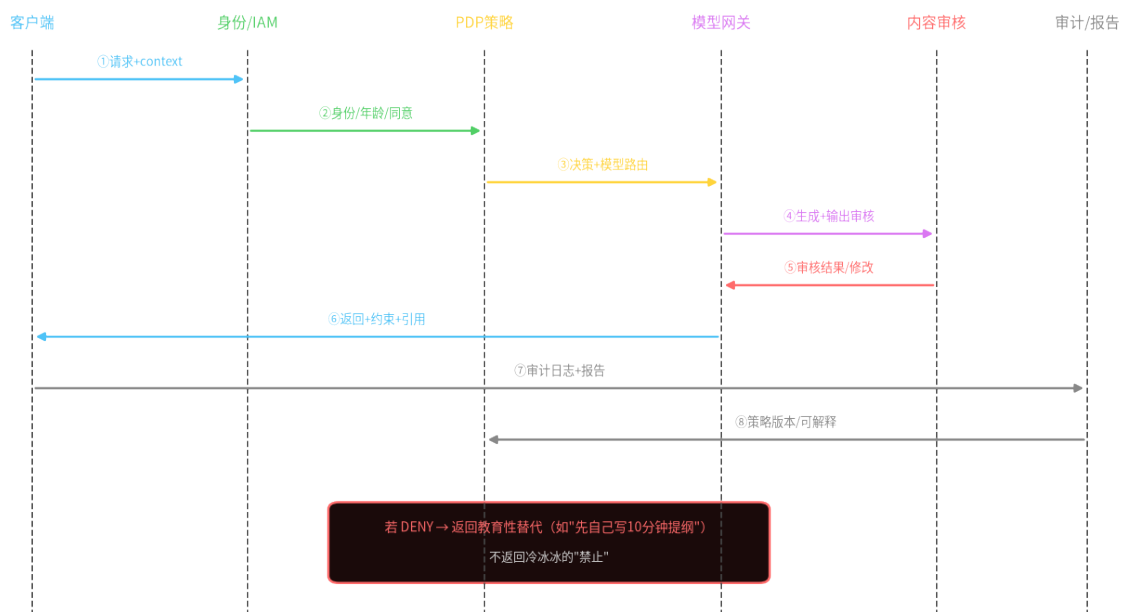


Figure 5 Core decision sequence eight steps: client request→identity authentication and age assertion→PDP policy evaluation→model gateway routing and capability matching→content safety input moderation→education orchestration (Socratic/pre-task gate)→output moderation and audit metering→return decision and educational alternative. Bypass: memory and learning state asynchronously updated before/after request, not blocking main decision chain. (Concept schematic)

- **PEP (Policy Enforcement Point)**: deployed at API gateway, client SDK, LMS plugin entry. Responsibilities: intercept all AI requests → collect context attributes → call PDP → execute decision (allow/deny/downgrade/add constraints) → record audit. PEP does not do policy judgment, only execution.
- **PDP (Policy Decision Point)**: independently deployed policy engine. Responsibilities: receive attributes → load policy version → evaluate → return decision. PDP stateless, horizontally scalable.
- **PAP (Policy Administration Point)**: admin backend, responsible for policy authoring, versioning, canary release, rollback.
- **PIP (Policy Information Point)**: provides attribute data sources (identity service, quota service, time service, learning state service).

This architecture references XACML (OASIS standard) PDP/PEP/PAP/PIP four-component separation, implemented in cloud-native scenarios with OPA/Rego or Casbin. **Policy decoupled from models** is the core principle—change model without changing education rules, change education rules without changing model.

4.5 Policy as Code

Classification policies written in declarative language (Rego or Casbin policy files), included in version control, supporting code review, testing, canary release, rollback. Example (Rego pseudocode):

```

# Lower primary (A1) student side default closes free generation
deny[msg] {
    input.subject.age <= 8
    input.subject.role == "student"
    input.model.requested_level >= "I3"
    msg := "Lower primary student side does not open free generation, please have teacher/parent proxy"
}

# Exam mode global disable
deny[msg] {
    input.scene.exam_mode == true
    msg := "Generative AI disabled in exam mode"
}

# Writing tasks must first submit outline
constraint[c] {
    input.task.type == "writing"
    input.subject.age <= 14
    c := "require_outline_first"
}

```

Policy files themselves have version numbers (V0–V3), each decision associated with policy version, supporting rollback to any historical version. This echoes UDOS evidence engineering's "version traceable" principle.

Chapter summary: decision function is pure, inputs four attribute categories, outputs nine enumerations + constraints; PDP/PEP separation ensures policy decoupled from business/models; Policy as Code makes policies versionable, testable, rollbackable; denial must give educational alternatives.

5 Ten-Layer System Architecture

One-sentence conclusion: the ten-layer architecture progresses layer by layer from access to admin backend, core is "classification policy center" as decision brain, "model gateway" as capability routing, "education orchestration" as learning process guard, "audit compliance" as full-chain traceability; each layer single responsibility, clear interface, independently replaceable.

5.1 Ten-Layer Architecture Overview

Table 5 Ten-layer system architecture definition

Layer	Name	Core responsibilities	Key components
①	Access layer	Multi-end access, protocol conversion, traffic entry	Web/App, school SSO, API, browser plugins, education software SDK
②	Identity and organization layer	Identity authentication, organization management, age assertion, guardian relationships	IAM (OIDC/SAML/Keycloak), organization/class/roles, guardians, age assertion service
③	Classification policy center	Policy decision, rule engine, version management	PDP/PEP, Policy as Code (OPA/Rego/Casbin), decision tables, canary/rollback
④	Content safety and grading	Input/output moderation, multimodal classification, crisis intervention	Multimodal moderation, classifiers (Llama Guard etc.), knowledge base, crisis intervention tickets
⑤	Model gateway	Model routing, prompt templates, RAG, tool calls, capability registration	LiteLLM routing, prompt templates, RAG whitelist knowledge base, tool whitelist, capability registration cards
⑥	Education orchestration layer	Task templates, Socratic guidance, answer-before-ask, process records	Task templates, SOCRATIC_HINT mode, pre-task gate, process evidence collection
⑦	Memory and learning state layer	Learning profiles, knowledge graph, mastery, cognitive load, originality detection	Learning profiles, knowledge graph, mastery models, cognitive load assessment, originality detection
⑧	Quota and time layer	Metering, quotas, rate limiting, clocks, class mode	Metering service, quota center, rate limiting, clock policy, class mode/exam mode
⑨	Audit and compliance layer	Logs, explainability, reports, appeals, data retention	Audit logs, decision explainability, parent/teacher reports, appeal channels, data retention/deletion
⑩	Admin backend	School/parent/teacher/regulator multi-view management	School admin, parent controls, teacher workbench, regulator reporting

5.2 Core Data Flow

Client request

- ① access layer (PEP intercept)
- ② identity organization (authentication + age assertion + guardian consent)
- ③ classification policy center (PDP evaluate → decision + constraints)
- ⑤ model gateway (route to corresponding I-level model + prompt templates + RAG + tools)
- ④ content safety (input moderation → generation → output moderation)
- ⑥ education orchestration (Socratic mode/pre-task gate/process records)
- ⑧ quota time (deduct quota/update duration)
- ⑨ audit compliance (record decision+input+output+version → generate report)
- return client (with decision_id + constraints + citations + audit)

Bypass: ⑦ memory and learning state asynchronously updated before/after request (mastery, cognitive load, originality), not blocking main decision chain.

5.3 Three Key Design Notes

Design one: model gateway capability registration card. Each model/tool/Agent has a "capability registration card," recording: intelligence level (I), supported content level (C), applicable minimum age (A), version (V), whether memory capable, whether network capable, whether can generate code/images/files, cost/token rate, known limitations. The model gateway selects models satisfying constraints per PDP decision—e.g. decision requires I2+C1+A2, gateway only routes to models satisfying these three conditions, not allowing "requested model is used." This is **capability routing** not simple API proxy.

Design two: education orchestration layer "pre-task gate". This is the platform's most core education difference point. Before students call AI, they must first submit own thoughts, outline, draft or answer—the system verifies "student has attempted" (via text input, handwriting photo recognition, or teacher confirmation), then allows AI intervention. AI default enters SOCRATIC_HINT mode, only giving questions, clues, checkpoints, not direct answers. Senior high can open "counter-view mode," but must mark evidence and sources. This design directly echoes 2025 Steering Committee guidelines "primary prohibits independent open-ended generation" policy red line, also embodies the education philosophy "AI is hint coach, not ghostwriting tool."

Design three: audit compliance layer "decision explainability". Each decision records: who (userId), when, what scenario, which model version, which policy version, input attribute snapshot, output decision, reasons, whether teacher-overridden, override reason. Parents/teachers/regulators can query full decision chain by decision_id. This echoes EU AI Act high-risk system "record keeping" obligations, also echoes UDOS evidence engineering's "three-layer consistency" principle.

5.4 Technology Selection

Table 6 Technology selection recommendations (all checked to official docs, access date 2026-09-21)

Layer	Recommended components	Alternatives	Selection rationale	Limitations/notes
Policy engine	OPA/Rego	Casbin, Drools	CNCF graduated (2021-01-29), native PDP/PEP support, Rego strong expressiveness	Rego steep learning curve; policy correctness needs testing
Identity	Keycloak (OIDC/SAML)	Auth0, school SSO integration	Open source, federated identity support, common in education	Needs integration with education student status systems
Gateway	Kong/APISIX/Envoy	—	PEP deployment point, rate limiting, authentication	Gateway itself becomes focus point
Model gateway	LiteLLM	Self-developed routing	Unified 100+ providers (vendor caliber), cost tracking, fallback	Enterprise SSO paid version; tool call support inconsistent across providers
Content safety	Llama Guard 3 + NeMo Guardrails + Presidio	Guardrails AI, self-developed classifiers	Llama Guard S1–S13 classification, NeMo orchestration, Presidio PII masking	Chinese education sensitive words need self fine-tuning; cannot replace human review
Knowledge/vector	pgvector/Milvus	—	RAG whitelist textbook library	Needs regular textbook version updates
Events/workflow	Kafka/Temporal	—	Async audit, quota deduction, retries	Increases system complexity
Data storage	Postgres/Redis/ClickHouse	—	Structured data/cache/audit log analysis	Minor data needs encryption + regional residency
Observability	OpenTelemetry/ELK	—	Full-chain tracing, log aggregation	—
Deployment	K8s	Privatized/hybrid cloud	Regional data residency, elastic scaling	Education privatized deployment needs operations capability consideration

Chapter summary: ten-layer architecture single responsibility, clear interface; three key designs (capability registration card, pre-task gate, decision explainability) are education scenario core differences; technology selection prioritizes mature open-source components, but Chinese education sensitive words need self fine-tuning, AI detection tools cannot serve as sole judgment basis.

6 Data Model and Key APIs

One-sentence conclusion: data model designed around "decision traceability," core entities User, Policy, Decision, AuditLog, ProcessEvidence; API with `/v1/ai/query` as core entry, all requests carry context, all returns carry decision_id, ensuring end-to-end auditability.

6.1 Core Data Model

Table 7 Core data entities and key fields

Entity	Key fields	Description
User	user_id, role, age, grade, org_id, class_id, status	Roles: student/parent/teacher/researcher/admin/regulator
AgeProfile	user_id, age_band (A0–A5), birth_date, age_assertion_method, guardian_verified	Age assertion methods: student status/parent account/government ID/self-declaration (low risk only)
GuardianConsent	consent_id, student_id, guardian_id, scope, granted_at, revoked_at, status	Revocable; scope limits data use
Role/Permission	role_id, permissions[], data_scope, final_assessment (bool)	final_assessment only teacher true
Org/Class	org_id, org_type (school/district), class_id, grade	Organization hierarchy
Policy	policy_id, name, description, version (V0–V3), status, code (Rego), created_by	Policy as code
PolicyVersion	version_id, policy_id, version_str, changelog, canary scope, created_at	Supports canary/rollback
Resource/Content	content_id, content_type, safety_level (S), topic_level (C), source_level (G), labels[], creator	Content labels
ContentLabel	label_id, content_id, label_type, value, confidence, label_version	Labels versioned
Model	model_id, name, provider, version, capability_level (I), cost_per_token, status	Model registration
ModelCapability	model_id, capability_card (json), min_age (A), max_content (C), tools[], limitations	Capability registration card
Session	session_id, user_id, scene, started_at, ended_at, duration, mode	Session
Request	request_id, session_id, prompt_hash, attachments_hash, task_type, cognitive_level (L), timestamp	Request (content hash, no original stored to protect privacy)
Decision	decision_id, request_id, policy_version, model_version, input_snapshot, output (json), reasons[], created_at	Core traceable entity
Quota	user_id, period, quota_type (token/calls/time/cost), limit, used, remaining	Quota
TimeWindow	user_id, date, window_type (T0–T4), allowed_hours, used_minutes, cooldown_until	Time
AuditLog	log_id, decision_id, actor, action, timestamp, metadata	Audit
Consent	consent_id, user_id, type, scope, granted_at, revoked_at	Consent management
Guardian	guardian_id, student_ids[], relationship, contact_verified	Guardian
ClassroomMode	class_id, mode (active/exam/locked), teacher_id, started_at, ended_at	Class/exam mode

Entity	Key fields	Description
Assessment	assessment_id, student_id, task_id, teacher_id, ai_assisted (bool), process_evidence_ids[], final_score, teacher_notes	Teacher final assessment, AI no final evaluation
ProcessEvidence	evidence_id, session_id, type (draft/outline/prompt/modification/citation), content_hash, timestamp, ai_usage_declaration	Process evidence: draft history/ prompts/modification trajectory/ citations
TeacherOverride	override_id, decision_id, teacher_id, original_decision, new_decision, reason, timestamp	Teacher can override but logged

6.2 Key APIs

Table 8 Key API definitions

Method	Path	Description	Key inputs	Key outputs
POST	/v1/ai/query	Core AI query entry	context (userId, role, age, scene, task, prompt, attachments, requested_l_level)	decision_id, allowed, mode, model, constraints, content, citations, hints, audit
POST	/v1/policy/evaluate	Policy evaluation (debug/audit)	input_attributes	decision, reasons, policy_version
POST	/v1/content/classify	Content classification	content_text/hash	safety_level (S), topic_level (C), source_level (G), labels
POST	/v1/quota/check	Quota check	userId, quota_type	remaining, limit
POST	/v1/quota/consume	Quota deduction	userId, quota_type, amount	new_remaining
POST	/v1/time/check	Time window check	userId, scene	allowed, current_window (T), remaining_minutes
POST	/v1/audit/log	Audit log write	decision_id, actor, action, metadata	log_id
POST	/v1/guardian/consent	Guardian consent grant/revoke	student_id, guardian_id, scope, action	consent_id, status
POST	/v1/teacher/override	Teacher override decision	decision_id, new_decision, reason	override_id
GET	/v1/report/guardian/{student_id}	Parent report	student_id, period	usage_summary, decisions, alerts, quota
GET	/v1/audit/decision/{decision_id}	Decision detail query	decision_id	full decision chain

6.3 /v1/ai/query Contract

Request must carry: user identity, age, role, class, scenario (class/homework/exam/home/project), task type, whether teacher supervised, whether parent consented, requested model capability level, content attachment hash, privacy flags.

Return must include: decision ID, whether allowed, education mode (direct answer/SOCRATIC_HINT/clues only/deny with alternative), citation sources, quota and time constraints, audit ID.

Design proposal • to be implemented [RESULT NEEDED]: the above APIs are design proposals, not yet implemented. Actual implementation needs adding: authentication middleware, rate limiting, error code specs, WebSocket streaming response support, versioned API migration strategy.

Chapter summary: data model designed around Decision core traceable entity, ProcessEvidence and TeacherOverride are education-specific entities; API with /v1/ai/query as core, all requests carry context, all returns carry decision_id; all design proposals • to be implemented.

7 Stage Strategy Matrix and Education Orchestration

One-sentence conclusion: the stage strategy matrix lands the ten-dimension code into each age band's specific permissions and constraints, numbers all strategy examples • configurable; the education orchestration layer via six mechanisms "pre-task gate, Socratic hints, process evidence, AI usage declaration, teacher final assessment, exam lock" ensures "AI is hint coach, not ghostwriting tool."

7.1 Stage Strategy Matrix

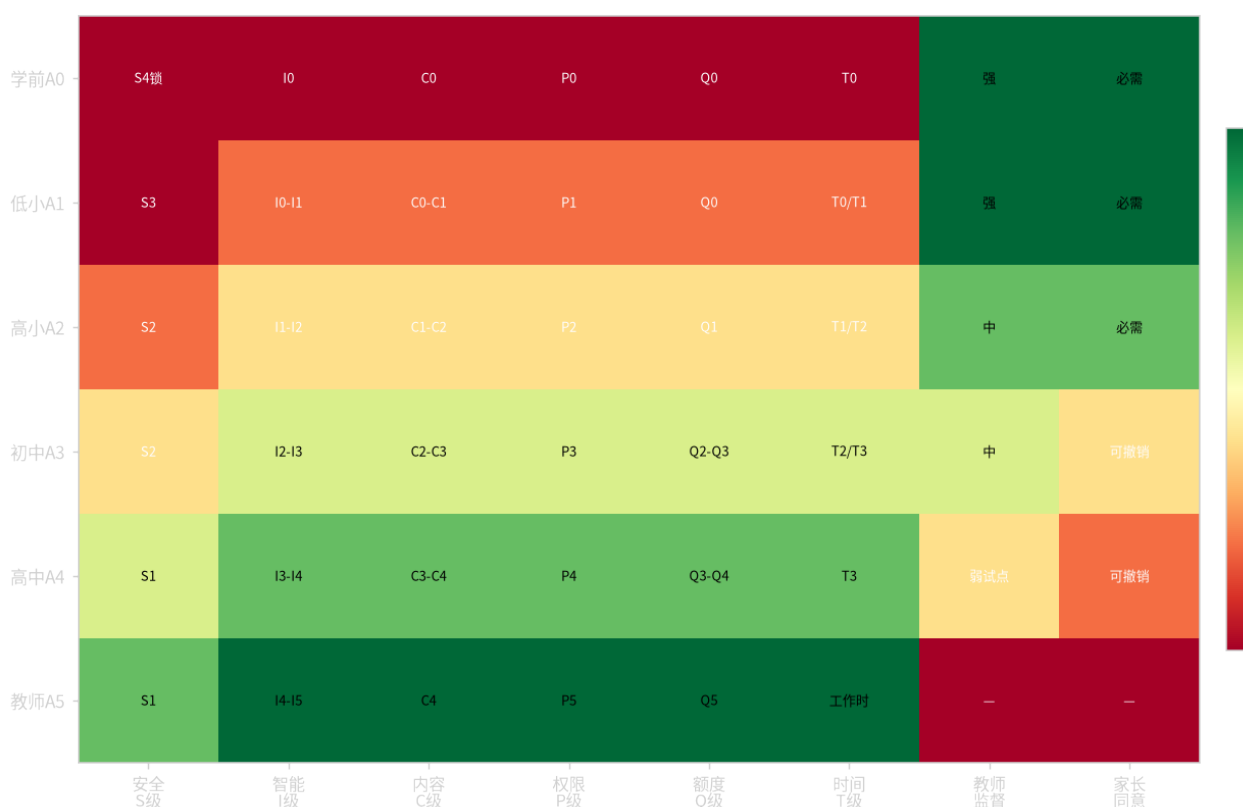


Figure 3 Stage X dimension strategy matrix (strategy example • configurable, not measured; greener higher openness). Preschool/lower primary locked, upper primary controlled Q&A, junior high

generation+citation+process, senior high open+critique+project, teachers full function but no final evaluation.

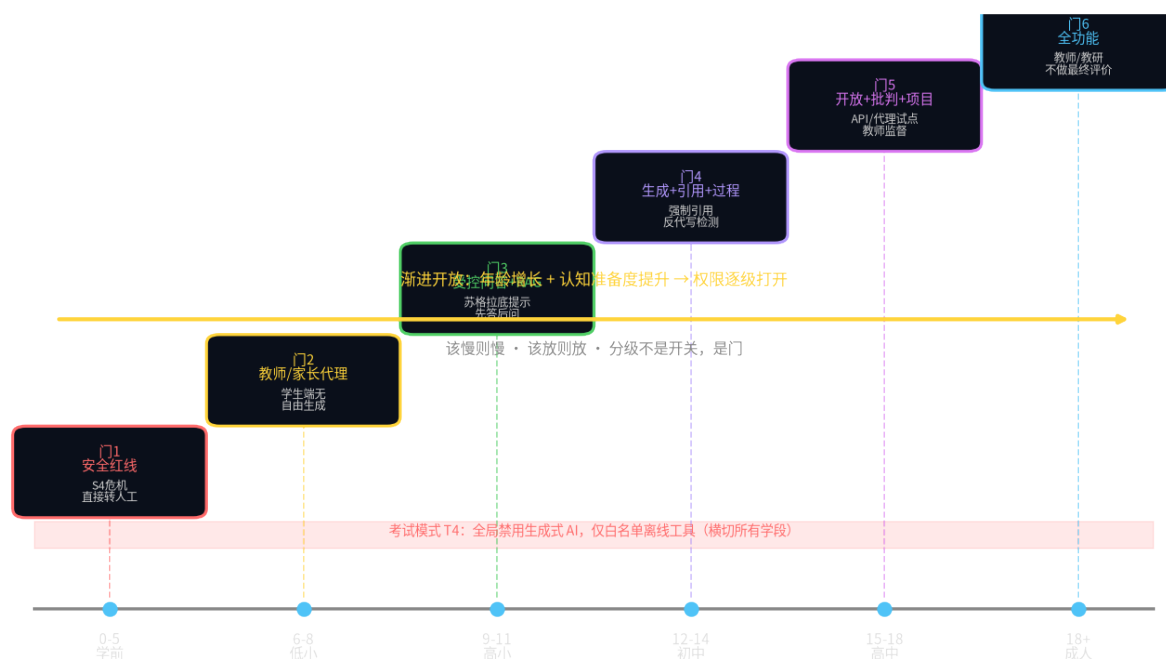


Figure 4 Progressive gate opening six gates: age axis A0→A5, permissions from "safety red line/teacher proxy" opened level by level to "full function/audit." Each gate corresponds to a default strategy set, adjustable by teacher/parent/regulator within authorized scope, but cannot skip safety red line. "Classification is not a switch, it is gates: slow when slow needed, open when open needed." (Strategy example • configurable)

Table 9 Stage strategy matrix (all quota/duration numbers strategy examples • configurable, not measured)

Stage	Age	Default permissions	Intelligence level (I)	Typical mode	Quota/time (example)	Supervision
Preschool A0	0-5	Student side no free generation	I0-I1	Teacher/parent proxy, whitelist content, stories/literacy	Q0, T0/T1	Strong
Lower primary A1	6-8	Student side no free generation	I0-I1	Teacher/parent proxy, whitelist Q&A	Q0, T0/T1	Strong
Upper primary A2	9-11	Limited Q&A + textbook RAG	I1-I2	Socratic hints, answer-before-ask, hint cards	10-20 calls/day, 15-20min	Parent consent
Junior high A3	12-14	Can generate but mandatory citation + process records	I2-I3	Debate, experiments, logic analysis, anti-ghostwriting detection	30-50 calls/day, 30min	Teacher supervision
Senior high A4	15-18	Open + critique + project + API teacher supervised pilot	I3-I4	AI literacy, counter-view, verification, project creation	80-200 calls/day, 60min	Teacher pilot, exam lock
Teacher A5	18+	Full function but no final evaluation	I4-I5	Lesson prep, question setting, grading assist, learning state analysis	Per organization quota	Audit
Exam scenario	any	Global generative AI disable	I0	Only whitelist offline tools (calculator/dictionary)	Q0, T4	Invigilation

Table 9 note: quota/duration numbers strategy examples • configurable, actual deployment should be adjusted by school/region per learning state, devices, network conditions, and verified through pilots. Low-age "no free generation" directly corresponds to 2025 Steering Committee guidelines "primary stage prohibit students independently using open-ended content generation functions."

7.2 Education Orchestration Six Mechanisms

Mechanism one: Pre-task Gate. Before students call AI, they must first submit own thoughts, outline, draft or answer. The system verifies "student has attempted" (via text input, handwriting photo recognition, or teacher confirmation), then allows AI intervention. This is the technical implementation of "think yourself first"—AI cannot replace the thinking process, only provide feedback after thinking.

Mechanism two: SOCRATIC_HINT. Low-age default mode, AI does not directly give answers, only questions, clues, checkpoints. E.g. student asks "how to do this math problem," AI replies: "First think what operation this involves? What known conditions are there? What can the first step be?" Senior high can switch to "counter-view mode," AI gives opposing argument, but must mark evidence and sources.

Mechanism three: Process Evidence. The system saves draft history, prompt records, modification trajectory, AI dialogue summaries, citation sources. Each process evidence carries content_hash (no original stored to protect privacy) and timestamp. Teachers can view students' complete learning process—this has more education value than "whether final assignment AI-generated," also more reliable than AI detection scores.

Mechanism four: AI Usage Declaration. Assignments automatically carry metadata: which model used, which paragraph used AI, what prompts, AI participation level (G0–G4). This corresponds to Labeling Measures + GB 45438 explicit/implicit label requirements, also cultivates students' academic integrity awareness.

Mechanism five: Teacher Final Assessment. AI can assist grading, give feedback, analyze learning state, but **final_assessment permission only teacher**—AI cannot directly decide student evaluation. Teachers can override AI decisions (TeacherOverride), but must log (override reason, time). This is the core embodiment of "human in loop."

Mechanism six: Exam Mode Lock. Exam scenarios global T4/Q0/I0, disable generative AI, only allow whitelist offline tools (calculator, dictionary). Class mode switched by teacher one-click, exam mode time window configured by school admin. This is the technical guarantee of evaluation fairness.

7.3 Denial Educational Alternative Examples

Scenario	System decision	Educational alternative reply
8-year-old requests "help me write essay"	DENY + TEACHER_ONLY	"At this stage first write an outline yourself for 10 minutes, then you can have teacher or mom/dad take a look."
10-year-old requests "give me answer directly"	ALLOW_WITH_GUARDRAILS + SOCRATIC_HINT	"First say what you think? What do you feel the first step should be? I can help check your thinking."
14-year-old writes essay without submitting outline	DENY + pre-task gate	"Please first submit your essay outline (at least 3 points), after submission I can help refine logic and wording."
Any AI request during exam	DENY + T4	"AI disabled in exam mode, please complete independently."
Senior high requests "help me write full paper"	ALLOW_WITH_GUARDRAILS + mandatory citation + process records	"I can help outline, find literature, check argument, but the paper needs you to write. Each AI-assisted paragraph auto-labeled."

Chapter summary: stage strategy matrix lands ten-dimension code into specific age bands, numbers all strategy examples • configurable; education orchestration six mechanisms (pre-task gate, Socratic hints, process evidence, AI usage declaration, teacher final assessment, exam lock) are the technical implementation of "AI is hint coach not ghostwriting tool"; denial must give educational alternatives.

8 Mapping with UDOS Governance System

One-sentence conclusion: this platform's ten-dimension classification and UDOS governance system are deeply isomorphic on five dimensions—I0–I5 align UDOS L0–L5 autonomy levels, Policy as Code aligns evidence engineering "rules before learning," TransferBundle least privilege aligns permission grading, BFT-lite multi-person review aligns teacher committee/human review gates, quality saturation law explains the governance ceiling of "generation scalable, teacher review not scalable."

8.1 Five-Fold Mapping

Table 10 Five-fold mapping between this platform and UDOS governance system

Dimension	This platform design	UDOS counterpart	Mapping meaning
Autonomy grading	Intelligence I0–I5 (no AI→retrieval→template→dialogue→tool proxy→multi-agent autonomous)	UDOS L0–L5 autonomy levels; industry paper SOP gates	Education scenarios limit high autonomy (low-age I0–I2), consistent with UDOS "higher autonomy, higher governance cost" judgment
Rules first	Content safety/admission uses deterministic rules as base, learning classifiers as supplement	Evidence engineering H-CSC v2: deterministic lexical predicates AUROC 0.865–0.982 beat learning encoders 0.621–0.744	Education safety red lines (self-harm/violence/illegal) must use deterministic rules, cannot rely on learning classifiers' probabilistic output
Least privilege	Permission grading P0–P5, RBAC+ABAC, final_assessment only teacher	TransferBundle scope least privilege field	Each AI interaction only grants minimum permissions needed for task, isomorphic with UDOS "handoff as contract" least privilege principle
Multi-person review	Teacher can override decisions, teacher committee reviews high-risk decisions, S4 crisis to human	BFT-lite three-level typed finality; review gates (5 human + 1 red team + 1 version)	High-risk/high-stakes decisions need multi-person review, single AI decision or single teacher decision both insufficient
Governance ceiling	Generation scalable, teacher review not scalable; AI handles standardized, teacher focuses individual differences	Quality saturation law (verification capacity ceiling: generation scalable, review not scalable, Little's Law/WIP=1)	Platform goal uses AI to scale "generation/feedback/analysis" capacity, freeing teachers from standardized work to focus individual judgment—but teacher review itself not scalable, this is governance hard ceiling

8.2 Rules Before Learning: H-CSC Evidence Implications for This Platform

UDOS Paper 06 H-CSC v2's core evidence: **deterministic lexical predicates AUROC 0.865–0.982 beat learning semantic encoders 0.621–0.744** (correct caliber after author voluntarily withdrew v1 "honestly retained advantage" claim). This has direct implications for this platform's content safety design:

- **Safety red lines (S3/S4) use deterministic rules:** self-harm, violence, illegal, minor privacy etc. red-line content must use deterministic rules (keywords, regex, pattern matching), cannot rely on learning classifiers' probabilistic output. Deterministic rules' advantages explainable, auditable, zero hallucination—exactly what education scenarios most need.
- **Learning classifiers as supplement:** content topic grading (C0–C4), age-appropriateness judgment, controversial content identification etc. "gray" decisions can use learning classifiers assist, but must give confidence, low confidence to human.

- **Learning encoders' failure boundary:** learning classifiers under out-of-distribution input, adversarial prompts, multilingual mixing scenarios performance significantly drops (H-CSC evidence shows learning encoders AUROC only 0.621–0.744). Education students will try various bypasses (homophones, character splitting, multilingual), deterministic rules more robust.

8.3 Quality Saturation Law: Why Teacher Final Assessment Cannot Be Omitted

UDOS Paper 07 quality saturation law's core judgment: **generation capability scalable (add more models/more compute), but verification/review capability not scalable (teacher time hard constraint)**, expressed via Little's Law as WIP=1 review throughput has ceiling.

This means for this platform:

- **AI can scale "feedback capacity":** auto-grade multiple choice, generate practice questions, analyze learning state data—these standardized work, AI can scale.
- **Teacher cannot scale "judgment capacity":** students' final evaluation, personalized feedback, psychological care, ethical judgment—these need human teachers, AI cannot replace.
- **Platform goal not "replace teacher with AI," but "free teacher from standardized work with AI,"** letting teachers have more time for what AI cannot do. This is the economic basis of "AI provides information and efficiency, human retains judgment and responsibility."

Chapter summary: this platform and UDOS governance system five-fold isomorphic (autonomy grading, rules first, least privilege, multi-person review, governance ceiling); H-CSC evidence proves safety red lines use deterministic rules, learning classifiers supplement; quality saturation law explains the economic reason teacher final assessment cannot be omitted.

9 Essence and Dross: Technology Selection Critique

One-sentence conclusion: not all "AI safety tools" suit education scenarios—LiteLLM/OPA/Presidio directly adoptable, Llama Guard/NeMo Guardrails need Chinese education scenario self fine-tuning, AI-generated content detection tools (GPTZero/Turnitin/ZeroGPT) have peer-reviewed evidence proving systematic misjudgment against non-English writers, absolutely cannot serve as sole academic discipline basis, age estimation AI has bias and privacy controversy, only auxiliary signal.

9.1 Essence: Components Suitable for Direct Adoption

1. LiteLLM as LLM gateway layer. Strips "which model, which key, how much cost, how to fallback when down" from business code. Education platforms can self-host without uploading student data to vendor telemetry; cost tracking and budget rate limiting highly valuable for school/region procurement. Limitations: enterprise SSO/SCIM paid version; different providers' tool call/streaming/JSON Schema support inconsistent (vendor caliber "140+ providers/1800+ models," mark vendor caliber when citing).

2. Presidio for PII masking before student data leaves domain. Before student chat/assignments sent to external large models, first identify names, student IDs, parent phones, school names etc. and replace with placeholders. This is the highest cost-performance layer in FERPA/COPPA/GDPR/PIPL compliance chain. Limitations: Chinese names, low-resource languages need self-training; masking ≠ anonymization (linkage attack risk); governance transition statement (2026 transition to data-privacy-stack) from third party, needs Microsoft official announcement confirmation [CITATION NEEDED].

3. OPA/Rego or Casbin as policy layer. Writing classification policies like "primary grades disable open chat, junior high only whitelist models, exam period prohibit generative AI" as code, PDP independently deployed, naturally aligned with EU AI Act high-risk obligations' needed "human supervision, record keeping" structure. OPA CNCF graduated (2021-01-29), Casbin multi-language support. Limitations: Rego steep learning curve; Casbin ABAC expressiveness weaker than OPA/Rego.

4. XACML/NIST RBAC/ABAC as architectural terminology and benchmark. No need to land full XACML (XML cumbersome, cloud-native mostly replaced by OPA), but citing ANSI/INCITS 359-2012 (RBAC) and NIST SP 800-162 (ABAC) can elevate "graded access control" from engineering slogan to standard language.

9.2 Components Needing Customization or Secondary Training

5. Llama Guard 3 / NeMo Guardrails / Guardrails AI as content safety guardrails. Llama Guard 3 covers S1–S13 total 13 safety risk categories (violent crimes, non-violent crimes, sex crimes, child sexual exploitation, defamation, professional advice, privacy, IP, indiscriminate weapons, hate, suicide/self-harm, sexual content, elections), but the taxonomy designed for **English adult safety scenarios**. Chinese education sensitive words (campus violence, minor inducement, exam leakage, beyond-syllabus content, value guidance) need self fine-tuning or self-written validators. Suggest using NeMo Guardrails as orchestration layer, Llama Guard 3 as one built-in risk signal, not sole judge. Limitations: rules can be bypassed via multi-turn dialogue; hallucination/fact checking relies on external scoring models.

6. Conformal prediction for "low confidence auto downgrade". Technical principle reliable (distribution-free uncertainty quantification, split conformal uses calibration set quantiles to give prediction sets, providing finite-sample coverage guarantee, ref Angelopoulos & Bates 2021, arXiv: 2107.07511), but LLM open-ended outputs have no unique correct labels. Suggest first piloting on multiple choice, knowledge point diagnosis, essay scoring etc. subtasks with clear answer sets, using split conformal to give "this AI answer confidence insufficient→to human/to more expensive model" fallback, not full scenario direct application. Limitations: relies on exchangeability assumption, coverage fails under data drift.

9.3 Dross: Known Failure Modes, Cannot Serve as Sole Judgment Basis

7. AI-generated content detection (GPTZero / Turnitin / ZeroGPT)—systematic bias with peer-reviewed hard evidence.

Liang, W., Yuksekogonul, M., Mao, Y., Wu, E., & Zou, J. (2023). "GPT detectors are biased against non-native English writers." **Patterns**, 4(7), 100779. PMID: 37521038.

Experiment: 7 mainstream GPT detectors × 91 non-native TOEFL essays from Chinese forum sources + 88 US Grade 8 essays from Hewlett Foundation ASAP dataset.

Conclusion: detectors systematically misjudge non-native English writing **as AI-generated**; near-perfect accuracy on US Grade 8 essays; simple prompt strategies both mitigate bias and can bypass detectors.

Meaning for education scenarios: **AI detection scores must not alone serve as academic misconduct discipline basis.** Credibility on Chinese writing lower (detectors mainly trained English). This platform at most treats detection scores as weak signal "suggest teacher review," and must combine triple evidence **human scoring, writing process logging (ProcessEvidence), student self-statement**. Third-party reports say Turnitin self-reports non-native misjudgment about 6–9%, native 1–4% (this number from Turnitin's own research second-hand retelling, before formal citation suggest checking Turnitin

official report [CITATION NEEDED]); Vanderbilt University 2023-08, Curtin University 2026-01 successively stopped using Turnitin AI detection (institutional decisions, citable as governance cases).

8. Facial age estimation / self-declared age gate—bypass and bias. Self-declared age easily bypassed (students casually fill an adult age); facial age estimation has bias and privacy controversy for minors, different skin colors, glasses/mask scenarios (ICO 2025–2026 progress documents already require platforms "beyond self-declaration"). Education scenarios if need distinguishing 13/14/16, should prioritize **government digital identity or parent account verification**, AI age estimation only as auxiliary signal, and cannot serve as sole admission basis.

9. "Open source = safe/compliant" intuition—open source $\neq$ compliance. Open-source components (Presidio, NeMo Guardrails, Llama Guard) themselves $\neq$ compliance; model training data licenses, whether student data enters fine-tuning, cross-border transfer, log retention cycles still need explicit declaration at Policy as Code layer. Llama Guard community license requires complying with Meta Llama community license terms.

Chapter summary: technology selection in three tiers—direct adoption (LiteLLM/OPA/Presidio), needs customization (Llama Guard Chinese fine-tuning/conformal pilot), cannot serve as sole basis (AI detection tools/age estimation AI); AI detection tools' systematic bias has Liang et al. 2023 peer-reviewed hard evidence, absolutely cannot serve as academic discipline sole basis.

10 Failure Boundaries, Falsifiability Conditions and Predictions

One-sentence conclusion: any classification governance platform has failure boundaries—private device bypass, third-party education software SDK integration, no absolute label standards, teacher burden, misjudgment/label bias, parent resource gap, policy fragmentation, over-blocking harms learning, cognitive readiness label ethical risk, formal compliance bypass; this chapter gives falsifiability conditions and specific predictions, before submission needs ≥ 30 seeds + 95% CI verification.

10.1 Failure Boundary List

Table 11 Failure boundaries and impact levels

#	Failure mode	Impact level	Description	Mitigation
1	Private device bypass	high	Students use personal phones/home computers to access AI products not connected to policy center, platform completely unable to control	School network layer control + parent end linkage + AI literacy education (teach students "what scenario use what")
2	Third-party education software SDK integration	high	School-purchased education software embeds AI, if not connected to policy center becomes governance blind spot	SDK/API mandatory policy center connection as procurement admission condition; education large model safety review mechanism
3	No absolute label standards	medium	Content grading (C0–C4), cognitive readiness (L0–L5) label boundaries fuzzy, different annotators may disagree	Labels versioned + multi-person annotation + consistency monitoring + low confidence to human
4	Teacher burden increase	medium	Teachers need review AI decisions, view process evidence, handle appeals, may increase workload	AI prioritizes standardized work (multiple choice grading, practice generation), teachers only review high-risk/high-stakes decisions; quality saturation law
5	Misjudgment/label bias	medium	Content classifiers, cognitive readiness assessment may bias specific groups (non-native, special education needs students)	Regular fairness audit + appeal mechanism + teacher review + no labels for ranking/tracking
6	Parent resource gap (digital divide)	medium	High-education/high-income parents better understand and use platform parent controls, low-resource families may not effectively supervise	Parent end simplified design + school provides unified default policy + no parent participation as student evaluation basis
7	Policy fragmentation	medium	Different schools/regions/teachers set different policies, causing cross-school student experience inconsistent	Regional baseline policy + schools can fine-tune on baseline + policy version management + best practice sharing
8	Over-blocking harms learning	medium	Over-conservative policy (e.g. all-stage generative AI ban) deprives students of AI literacy opportunity, contradicts MOE policy	Progressive opening principle + regular effect evaluation + student/teacher feedback channels
9	Cognitive readiness label ethical risk	high	L0–L5 labels if used for ranking, tracking, predicting future, cause label stigma and self-fulfilling prophecy	Only internal adaptation labels, prohibit external display/ranking/tracking/future prediction; appealable resettable; EU AI Act high-risk constraints
10	Formal compliance bypass	medium	Students/teachers may via "first submit junk outline then pass pre-task gate" etc. formally satisfy process, substantively still rely on AI	Process evidence quality assessment + teacher spot check + AI usage declaration + academic integrity education

10.2 Falsifiability Conditions

This platform's core design claims can be falsified via the following conditions:

1. **"Pre-task gate improves learning effect" falsifiable:** if A/B tests show student groups using pre-task gate **not outperform** (or underperform) control groups directly using AI on learning outcome tests, then this mechanism's education value is falsified. [RESULT NEEDED]
2. **"Ten-dimension grading superior to single age grading" falsifiable:** if comparison experiments show ten-dimension grading **not significantly superior** to simple age grading on user satisfaction, learning effect, safety incident rate, then ten-dimension complexity cost not worthwhile. [RESULT NEEDED]
3. **"Deterministic rules superior to learning classifiers for safety red lines" falsifiable:** if on education scenario real data, learning classifiers' AUROC **significantly higher** than deterministic rules (opposite H-CSC evidence), then rules-first design needs correction. [RESULT NEEDED]
4. **"Teacher final assessment cannot be omitted" falsifiable:** if AI auto evaluation on specific tasks (e.g. multiple choice grading) consistency with teacher evaluation **exceeds** inter-teacher consistency, then this task can be independently completed by AI, no teacher final assessment needed. [RESULT NEEDED]
5. **"Progressive opening superior to full opening/full ban" falsifiable:** if long-term tracking shows progressive opening group's AI literacy and academic performance **not superior** to full opening group, and safety incident rate **not lower** than full opening group, then progressive opening returns don't hold. [RESULT NEEDED]

10.3 Predictions

Table 12 Verifiable predictions and scoring time points

#	Prediction	Scoring time	Verification method	Current status
P1	Education AI classification platforms will before 2027 become standard in China primary/secondary informatization procurement (like current firewalls/online behavior management)	2027-06	MOE/provincial procurement catalog statistics	Prediction • to verify
P2	"Primary prohibits independent open-ended generation" policy red line will in 2026–2027 be refined by more regions into executable technical standards	2027-06	Local education department document search	Prediction • to verify
P3	AI-generated content detection tools will before 2027 be stopped by more universities (after Vanderbilt/Curtin), due to systematic bias evidence accumulation	2027-12	University announcements/media report statistics	Prediction • to verify
P4	China will before 2027 issue education AI classification dedicated technical standards or industry norms (extending GB 45438 to education scenarios)	2027-12	SAC/MOE standard release	Prediction • to verify
P5	Cognitive readiness label ethical controversy will before 2027 become public discussion topic, driving "prohibit using AI labels to track students" policy	2028-06	Media/policy document search	Prediction • to verify
P6	This platform's pre-task gate in A/B tests will show learning effect improvement (effect size $d > 0.2$)	6 months after pilot	Randomized controlled experiment, ≥ 30 seeds + 95% CI	[RESULT NEEDED]

Table 12 note: P1–P5 trend predictions, P6 experiment prediction. All predictions are design proposal inferences, not yet verified. Before submission needs ≥ 30 seeds + 95% CI verification of experiment predictions.

Chapter summary: among ten failure boundaries, private device bypass and third-party SDK integration highest impact; five falsifiability conditions clarify when design claims need correction; six predictions (with scoring time) provide roadmap for subsequent verification. All experiment predictions marked [RESULT NEEDED].

11 Compliance Checklist and Landing Roadmap

One-sentence conclusion: the compliance checklist translates regulatory requirements into technical actions platform must implement, landing roadmap in five phases (standard labels→MVP→pilot schools→model gateway + education orchestration→scale assessment), each phase with clear deliverables and verification standards, all design proposals • to be implemented.

11.1 Compliance Checklist

Table 13 Regulatory requirements → platform technical action mapping

Regulatory requirement	Source	Platform technical action	Implementation layer
Classified graded supervision	Generative AI Measures Article 3	Ten-dimension classification code + PDP decision	③ policy center
Prevent minor over-reliance addiction	Generative AI Measures Article 10	Time grading T0–T4 + quota grading Q0–Q5 + cooldown mechanism	⑧ quota time
Generated content labeling	Generative AI Measures Article 12 + labeling measures + GB 45438	Explicit labels (AI usage declaration) + implicit labels (metadata/watermark) + content source grading G0–G4	④ content safety + ⑥ education orchestration
Algorithm filing	Generative AI Measures Article 17 + algorithm provisions Article 24	Algorithm filing material preparation + policy version management	③ policy center + ⑨ audit
Under-14 guardian consent	PIPL Article 31 + children provisions	Age assertion + guardian consent process + revocable + special rules	② identity organization
Time/permission/consumption management	Minor Protection Regulations	Time grading + permission grading + quota grading + class/exam mode	⑧+⑨
Personal information minimum necessary	PIPL + Generative AI Measures Article 11	Data minimization collection + PII masking (Presidio) + content hash no original	②+④+⑦
Complaint reporting mechanism	Generative AI Measures Article 15	Appeal channel + teacher/parent review + decision queryable	⑨ audit compliance
Primary prohibits independent open-ended generation	2025 Steering Committee guidelines	A0–A1 student side no free generation + TEACHER_ONLY routing	③ policy center
Classify grade to determine safety protection standards	2026 MOE "AI+Education" Action Plan	Ten-dimension grading + safety grading S0–S4 + education large model safety review	Full platform

11.2 Landing Roadmap

Table 14 Five-phase landing roadmap (design proposal • to be implemented)

Phase	Name	Core deliverables	Verification standards	Estimated period
Phase 0	Standards and labels	Ten-dimension code specs, content label system, model capability registration card templates, policy DSL specs	Label consistency $\geq 85\%$ (multi-person annotation); policy files pass syntax check	1–2 months
Phase 1	MVP	Identity organization + policy center (PDP/PEP) + safety review + quota time + audit logs	Decision latency $< 200\text{ms}$ (p95); safety red line interception 100% (deterministic rules); audit log completeness 100%	2–3 months
Phase 2	Pilot schools	1–2 pilot schools (upper primary + junior high), teacher/parent console, stage strategy matrix landing	Teacher satisfaction $\geq 7/10$; safety incident rate $< 1/1000$ requests; student AI literacy pre/post improvement	3–6 months
Phase 3	Model gateway + education orchestration	LiteLLM routing + RAG textbook library + Socratic mode + pre-task gate + process evidence + AI usage declaration	Pre-task gate completion $\geq 80\%$; process evidence completeness $\geq 90\%$; teacher final assessment coverage 100%	3–6 months
Phase 4	Scale assessment and regulator reporting	Multi-school/region rollout; fairness audit; effect evaluation; parent reports; regulator reports; falsifiable prediction verification	Fairness audit no significant group bias; falsifiability condition verification complete; regulator reports generated on time	6–12 months

Table 14 note: all periods and verification standards design proposals • to be implemented [RESULT NEEDED]. Actual period depends on school cooperation, device conditions, teacher training progress.

11.3 Minimum Viable Verification Path

If resources limited, suggest prioritizing the following three minimum experiments:

- 1. Pre-task gate effect experiment:** in one junior high's two parallel classes (random assignment), one class uses pre-task gate + Socratic hints, one directly uses AI. After 6 weeks compare writing scores, AI reliance, student satisfaction. ≥ 30 seeds + 95% CI. [RESULT NEEDED]
- 2. Deterministic rules vs learning classifiers safety red line comparison:** on education scenario real data (including student bypass attempt cases), compare deterministic rules and learning classifiers' AUROC, false positive rate, false negative rate. Verify whether H-CSC evidence holds in education scenarios. [RESULT NEEDED]
- 3. Stage strategy matrix teacher acceptance survey:** distribute questionnaires to pilot school teachers, assess each stage strategy's rationality, operability, burden. Collect teacher revision suggestions, iterate strategy matrix. [RESULT NEEDED]

Chapter summary: compliance checklist maps 10 regulatory requirements to specific technical actions and implementation layers; landing roadmap in five phases, each phase with deliverables and verification standards; minimum viable verification path prioritizes pre-task gate effect, rules vs learning classifiers, teacher acceptance three experiments. All design proposals • to be implemented.

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5. UDOS v7.7.2 Paper 13. Evidence Engineering and Research Methods: three-level evidence grading, three-state ledger, revision traceability.

Evidence Ledger (EVIDENCE LEDGER v777 summary)

Full evidence ledger see same directory `EVIDENCE_LEDGER_v777.md`. This summary lists three-state distribution and key fact corrections.

Three-state distribution: Chinese regulations 14 items— verified 10,  partially verified 4 (2024 primary/secondary AI education notice doc no., 2026 "AI+Education" Action Plan five-unit full names

and doc no., 2025 nine-department digitalization opinion doc no., 2025 Steering Committee guidelines no order no. as guidance document), ■ no evidence 0. International systems 7 items—✓ verified 5, ● partially verified 1 (FERPA DOE official guidance), ■ no evidence 1 (Australia OSA first-hand text [CITATION NEEDED]). Technical components 11 items—✓ verified 8, ● partially verified 3 (Guardrails AI GitHub link, Presidio governance transition, star count/version no. not real-time pulled).

Key fact corrections: 1. Algorithm Recommendation Management Provisions by **four departments** Order No. 9 (CAC, MIIT, MPS, SAMR), not seven. 2. AI-Generated Synthetic Content Labeling Measures by **four departments** (including NRTA), doc no. Guoxinban Tongzi [2025] No. 2, not CAC alone. 3. GB 45438-2025 and Labeling Measures same day (2025-09-01) effective, transition period 6 months. 4. EU AI Act education scenario emotion recognition by Art.5(1)(f) **directly prohibited**, not high risk. 5. AI detection tools' systematic bias against non-English writers has Liang et al. 2023 Patterns peer-reviewed hard evidence, not rumor.

No-evidence disclosure: - Australia Online Safety Act 2021 first-hand text not retrieved this time [CITATION NEEDED]. - OPA/Casbin/Presidio/LiteLLM GitHub star counts and latest release version numbers not real-time pulled this time, formal draft needs supplement [CITATION NEEDED]. - DOE "Designing for Education with AI" (2024-07) official original PDF link not directly obtained this time [CITATION NEEDED]. - Turnitin self-reported misjudgment (non-native 6–9% vs native 1–4%) second-hand retelling, needs official report check [CITATION NEEDED]. - GDPR Art.8 member state consent age current values (FRA 2017 mapping older), before formal publication needs comparing each country DPA official site [CITATION NEEDED].

This paper is UDOS supplemental volume v7.7.7 P38, independent supplemental volume, not merged into v7.7.2's 13-paper master volume. Full text contains no new experiments; all design proposals marked "design proposal • to be implemented," quota/duration numbers marked "strategy example • configurable," external regulations and tools all with official sources and three-state marking. Before submission needs ≥ 30 seeds + 95% CI verification of experiment predictions.